Program 7 : Working with Selenium: Maven Project To Connect all the above 3 websites To Web Servers Using Selenium Dependency on Mozilla Firefox.

Step 0: Installation of Mozilla Firefox

- Download Gecko Driver (via terminal)

Run wget https://github.com/mozilla/geckodriver/releases/latest/download/geckodriver-v0.34.0-linux64.tar.gz

- Extract

tar -xvzf geckodriver-v0.34.0-linux64.tar.gz

- Move to system path

sudo mv geckodriver /usr/local/bin/

- Give permission

sudo chmod +x /usr/local/bin/geckodriver

- Verify installation

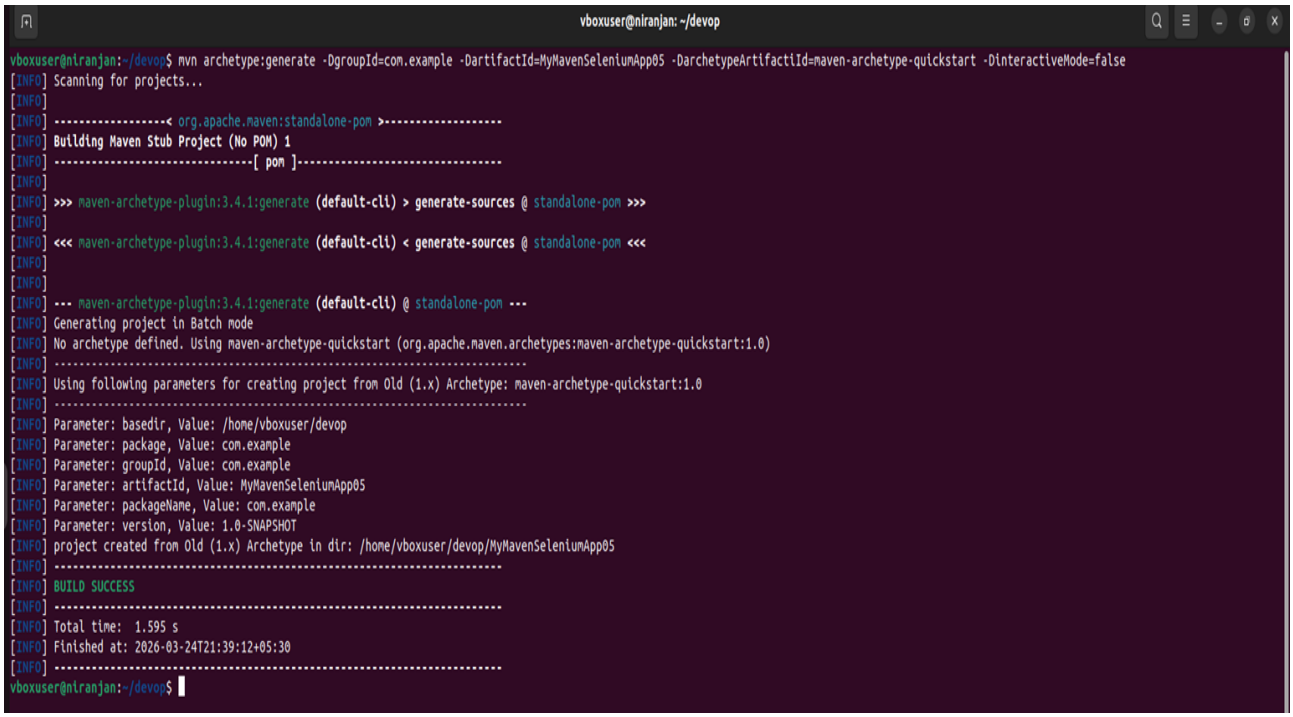
geckodriver –version

- **Install Firefox**

sudo apt update
sudo apt install firefox -y

Step 1: create a maven project on terminal by typing the command **mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp05 -DarchetypeArtifactiId=maven-archetype-quickstart -DinteractiveMode=false**

Step 2 : edit the pom.xml file and add the dependencies



The pom.xml has the code

Open Save

pom.xml
~/Devops/MyMavenSeleniumApp05

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>MyMavenSeleniumApp05</artifactId>
  <packaging>jar</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>MyMavenSeleniumApp05</name>
  <url>http://maven.apache.org</url>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
    <dependency>
    <groupId>org.seleniumhq.selenium</groupId>
    <artifactId>selenium-java</artifactId>
    <version>4.41.0</version>
    <scope>compile</scope>
</dependency>
  </dependencies>
  <!-- Build: Configuring plugins and build settings -->
<build>
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.7</source>
<target>1.7</target>
</configuration>
</plugin>
```

Open pom.xml ~/Devops/MyMavenSeleniumApp05 Save

```xml
<plugin>
            <groupId>org.apache.maven.plugins</groupId>

            <artifactId>maven-jar-plugin</artifactId>
            <version>3.0.1</version>
           <configuration>
                <archive>
                    <manifestEntries>
                        <Main-Class>com.example.App</Main-Class>
                    </manifestEntries>
                </archive>
            </configuration>


        </plugin>
        <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-shade-plugin</artifactId>
      <version>3.5.0</version>
      <executions>
        <execution>
          <phase>package</phase>
          <goals>
            <goal>shade</goal>
          </goals>
          <configuration>
            <transformers>
              <transformer implementation="org.apache.maven.plugins.shade.resource.ManifestResourceTransformer">
                <mainClass>com.example.App</mainClass>
              </transformer>
            </transformers>
          </configuration>
        </execution>
      </executions>
    </plugin>

</plugins>
</build>
</project>
```

Step 3: Edit the App.java

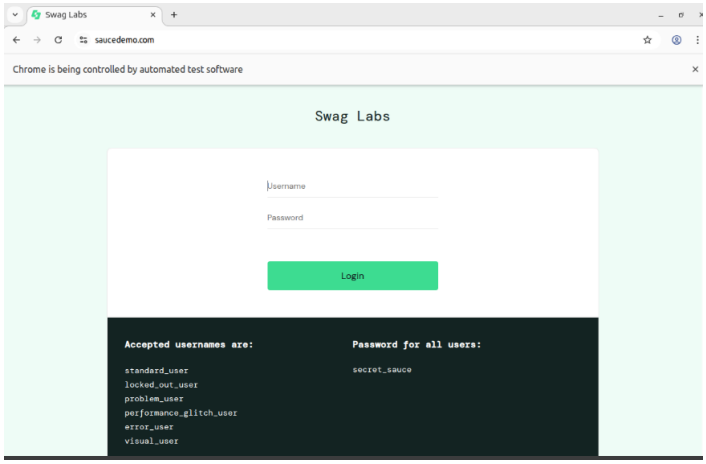
Open *App.java ~/Devops/MyMavenSeleniumApp05/src/main/java/com/example Save

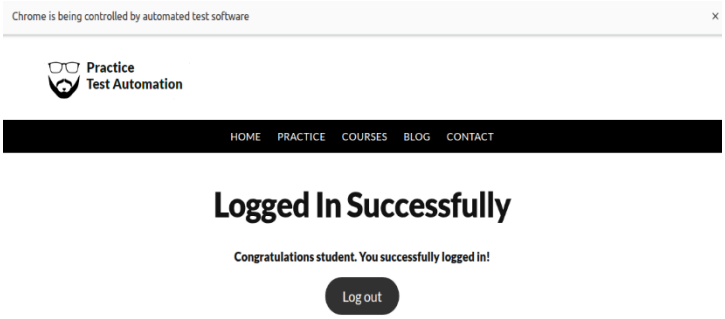
```java
package com.example;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.WindowType;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.interactions.Actions;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
import java.time.Duration;
public class App {

    public static void main(String[] args) throws InterruptedException {

        FirefoxOptions options = new FirefoxOptions();

        WebDriver driver = new FirefoxDriver(options);

        driver.manage().window().setSize(new org.openqa.selenium.Dimension(1920,1080));

        WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(15));

        Actions actions = new Actions(driver);

        driver.get("https://www.saucedemo.com/");

        driver.findElement(By.id("user-name")).sendKeys("standard_user");

        driver.findElement(By.id("password")).sendKeys("secret_sauce");

        driver.findElement(By.id("login-button")).click();

        System.out.println("SauceDemo login successful");

        driver.switchTo().newWindow(WindowType.TAB);

        driver.get("https://automationexercise.com/products");
```

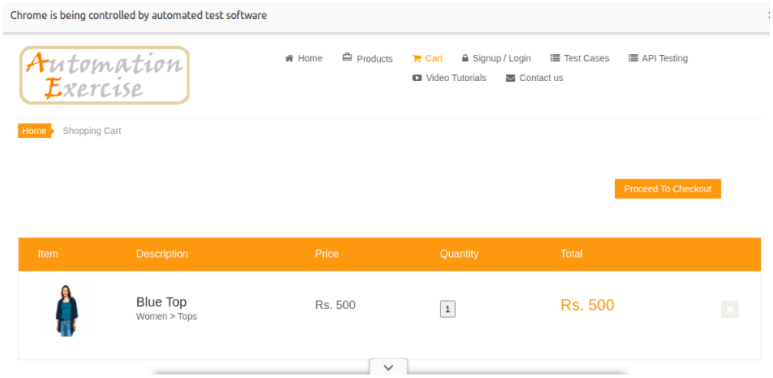
Open *App.java ~/Devops/MyMavenSeleniumApp05/src/main/java/com/example Save

```java
        ((org.openqa.selenium.JavascriptExecutor) driver)
                .executeScript("arguments[0].scrollIntoView({block:'center'});", product);

        ((org.openqa.selenium.JavascriptExecutor) driver)
                .executeScript("arguments[0].click();", product);

        WebElement viewCart = wait.until(
                ExpectedConditions.elementToBeClickable(
                        By.cssSelector("#cartModal a[href='/view_cart']")
                )
        );

        ((org.openqa.selenium.JavascriptExecutor) driver)
                .executeScript("arguments[0].scrollIntoView(true);", viewCart);

        ((org.openqa.selenium.JavascriptExecutor) driver)
                .executeScript("arguments[0].click();", viewCart);

        System.out.println("Automation Exercise product added to cart");

        driver.switchTo().newWindow(WindowType.TAB);

        driver.get("https://practicetestautomation.com/practice-test-login/");

        Thread.sleep(2000);

        driver.findElement(By.id("username")).sendKeys("student");

        driver.findElement(By.id("password")).sendKeys("Password123");

        driver.findElement(By.id("submit")).click();

        System.out.println("Practice Test Automation login successful");

        Thread.sleep(5000);

        driver.quit();
    }
}
```

Step 4:build and run the maven application
To execute type the command **<u>mvn exec:java -Dexec.mainClass="com.example.App"</u>**
it opens all the 3 websites automatically and registers using username and password.

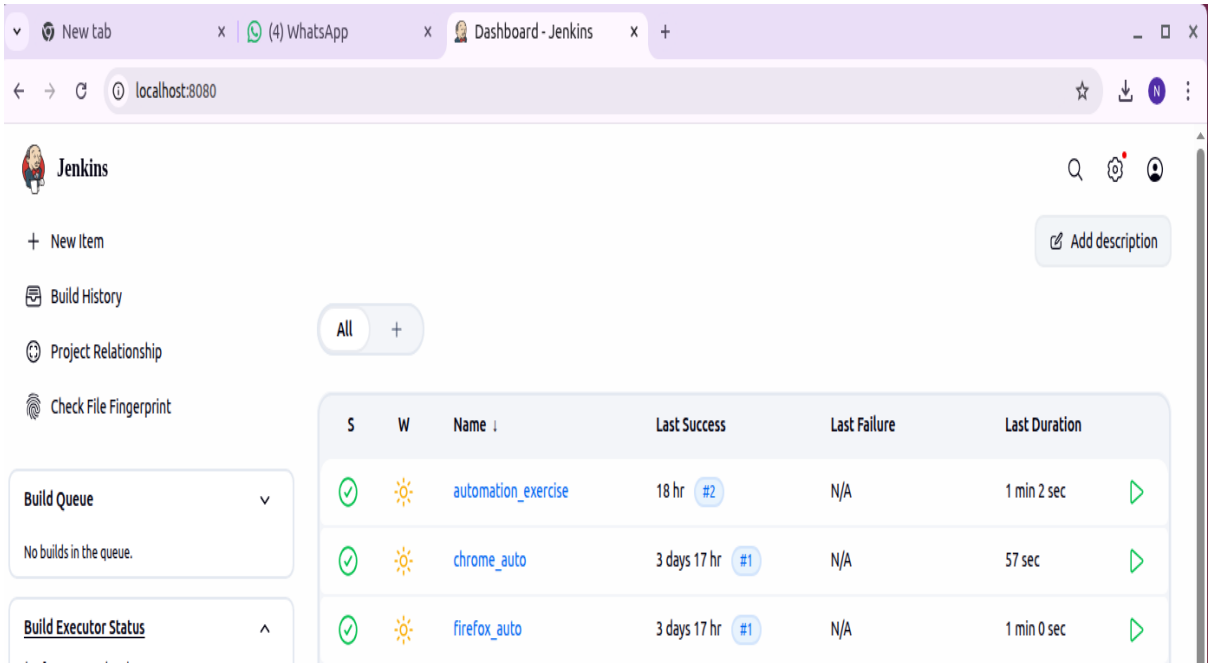




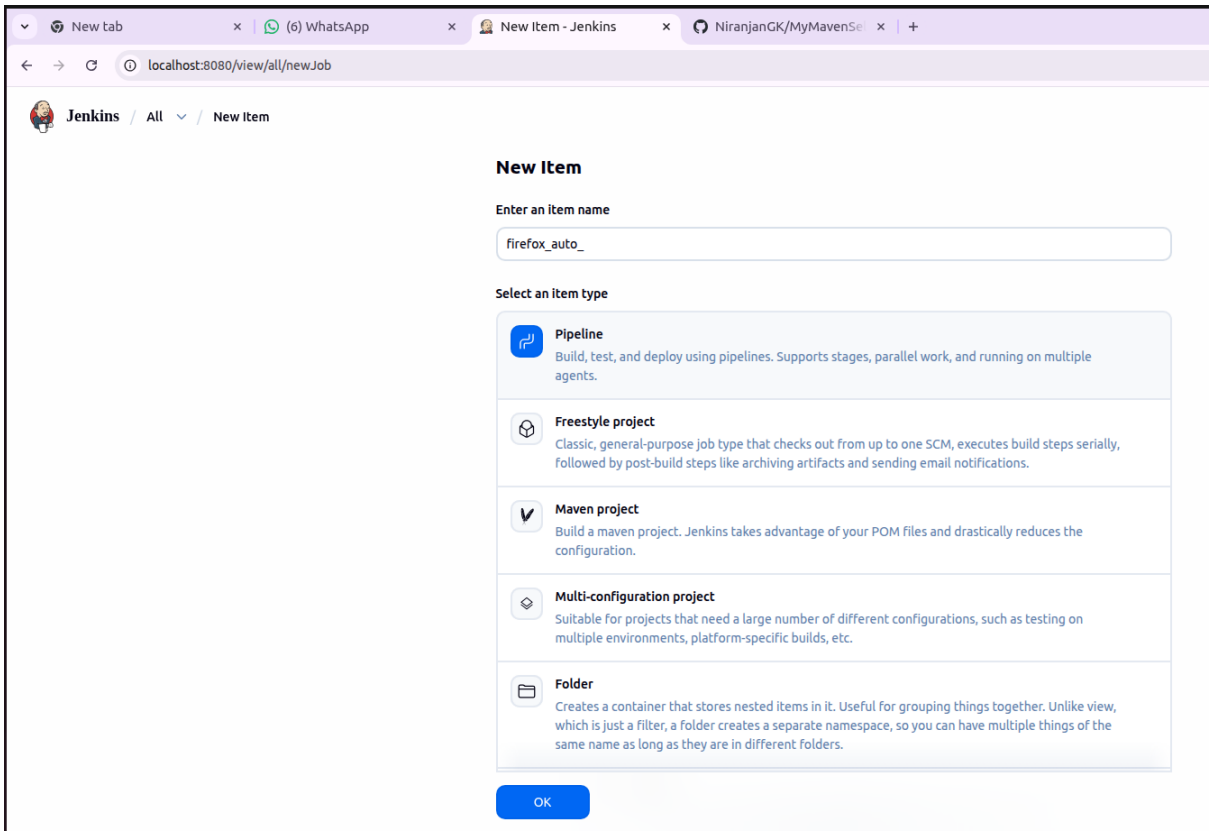


Jenkins Pipeline for MyMavenSelenium05

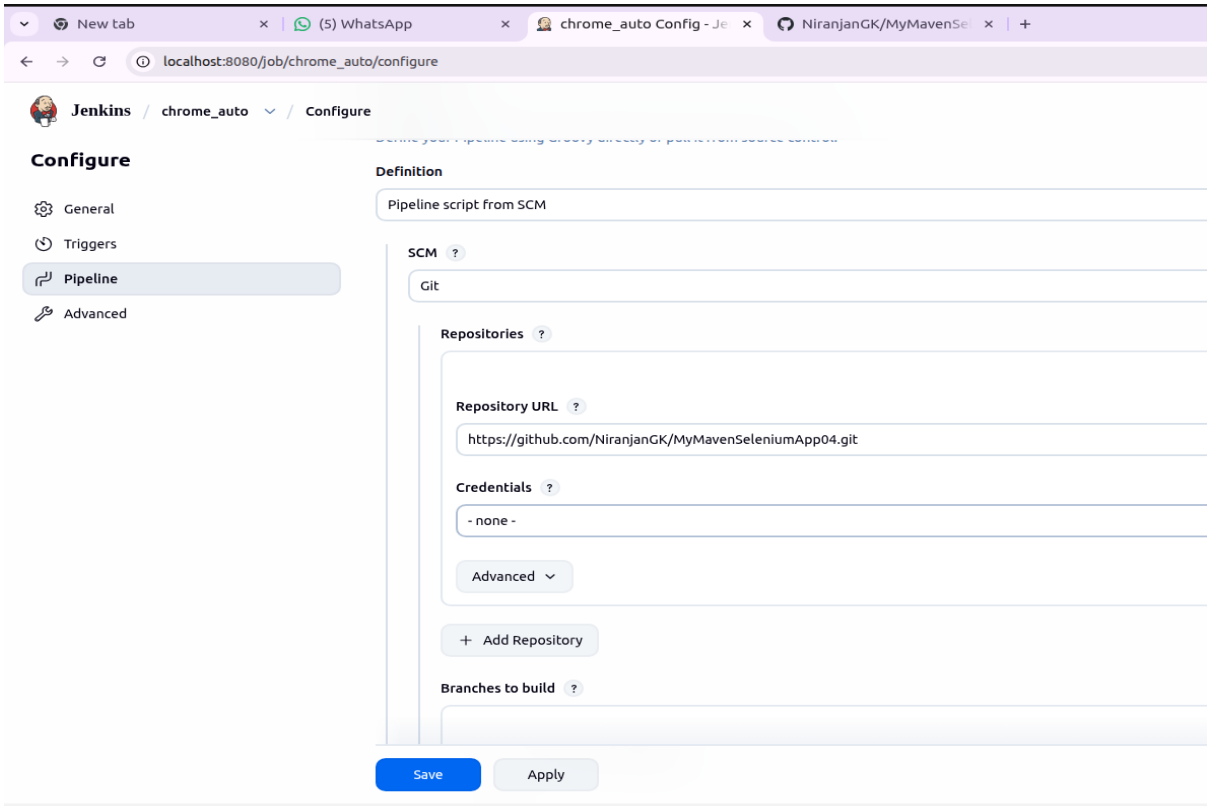
Step 1: Open the Jenkins Dashboard to create a new item



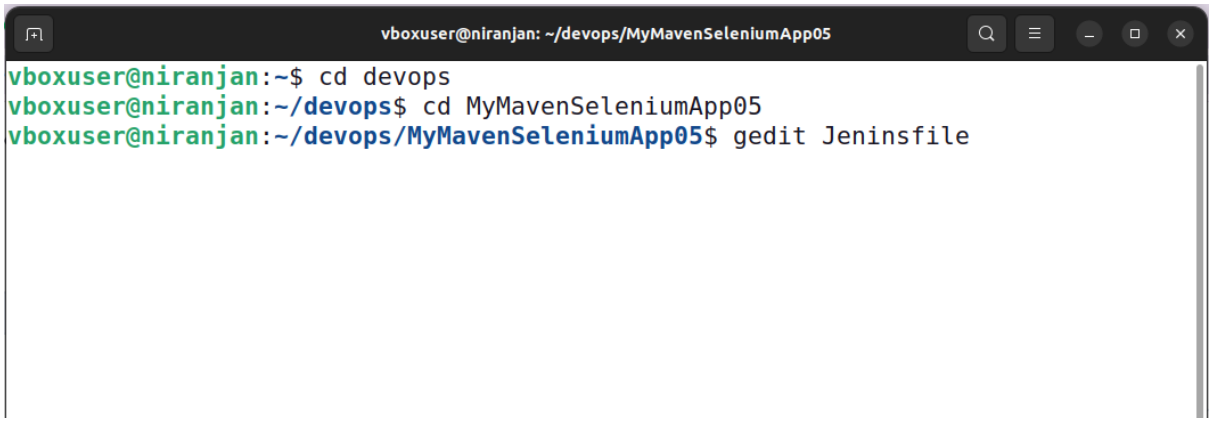
Step 2: Click on New Item to create an New Item and give the name as "firefox_auto" and Item type as Pipeline and click OK



Step 3: Configure the settings set script as “Pipeline Script from SCM” and SCM as “Git”. paste the Repository URL of the project given in GitHub and choose Credentials as “GitHub Credentials” which was created earlier.



Step 4: Now in Terminal, go to directory created for MyMavenAppSelenium03 and create an new file called “Jenkinsfile”



Step 5: Paste the available code inside the Jenkinsfile

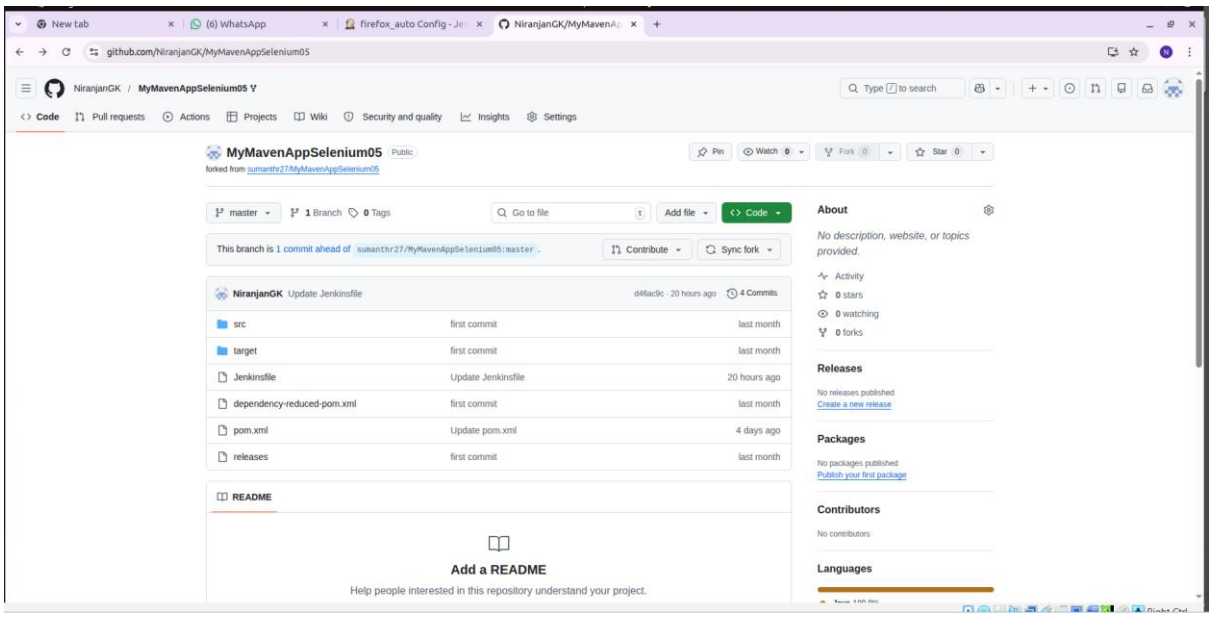
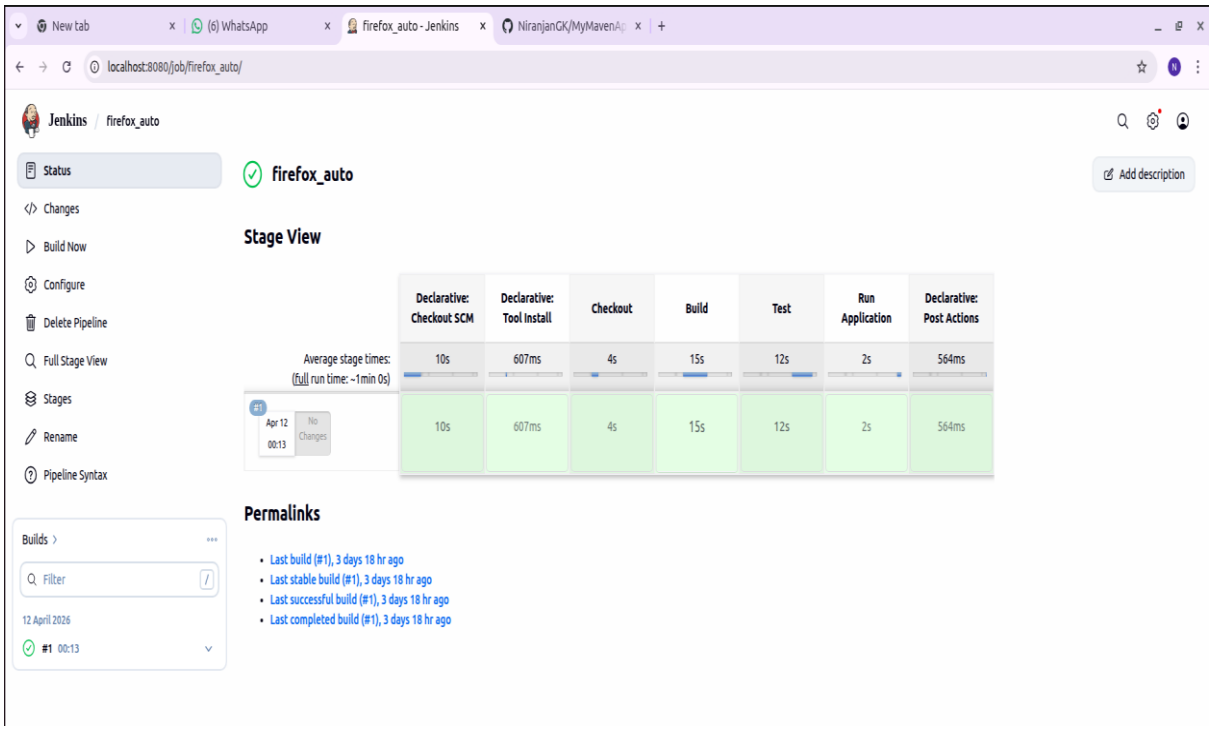
```
pipeline {
    agent any   // Run on any available agent


    tools {
        maven 'Maven'   // Must match name configured in Jenkins (Global Tool
Config)
    }


    stages {


        stage('Checkout') {
            steps {
                // Recommended if repo is configured in Jenkins
                checkout scm


                // OR use this if not configured:
                // git url:
'https://github.com/NiranjanGK/MyMavenSeleniumApp05.git', branch: 'master'
            }
        }


        stage('Build') {
            steps {
                sh 'mvn clean package'
            }
        }
```

```
        stage('Test') {
            steps {
                sh 'mvn test'
            }
        }
    }

    post {
        always {
            echo 'Pipeline execution completed'
        }
        success {
            echo 'Build SUCCESS '
        }
        failure {
            echo 'Build FAILED '
        }
    }
}
```

Step 6: We need to update pom.xml also in order to change the versions of source and target from 1.7 to 1.8

```
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>
```

Step 7: Now we need to Push changes and commit by pushing the necessary "Jenkinsfile" to the respective Repo



Step 8: Now go to Jenkins and Build the Pipeline and thus check the Stage view for the successful execution.